Post Graduate Common Entrance Test - 2015
Master Of Computer Applications

Time : 2hrs | Version Code :- A4 | Max Marks : 100

PART - 1

Each Question Carry One Mark(60 x 1 = 60)

1. A Microsoft Windows is a(an)
a. operating system
b. word processing
c. database program
d. graphic program

2. The Basic Input/Output System (BIOS) resides in
a. RAM b. ROM
c. CPU d. Memory Cache

3. A collection 8 bits is called
a. Byte b. Record
c. Word d. Nibble

4. In the binary number 110,101, the fractional part has the value
a. 0.625 b. 2^4
c. 2^{4-1} d. None of the above

5. The binary representation of the hexadecimal 'C3' is
a. 1111 b. 110011
c. 110001 d. 11000011

6. Which of the following memory is volatile?
a. RAM b. ROM
c. EPROM d. PROM

7. A four bit number is given as 1001, Its is complement is
a. 1001 b. 11001
c. 0110 d. 0101

8. Add$(1011010)_2$ and $(10110)_2$
a. 1010101 b. 1110101
c. 1110000 d. 1101010

9. Multiply binary number 1011 by 110
a. 1101010 b. 1011010
c. 1000010 d. 1011111

10. BCD is
a. Binary Coded Decimal
b. Bit Coded Decimal
c. Binary Coded Digit
d. Bit Coded Digit

11. If the decimal point is placed to the first significant digit, then it is called as
a. Orthogonal b. Normalised
c. Determinate d. None of the above

12. The first part of floating point represents a signed fixed point number called
a. Exponent b. Digit
c. Number d. Mantissa

13. In computer system, memory management operations are performed by
a. User b. Hardware
c. Programmer d. Operating System

14. A computer program that converts an entire program inot Machine Language is called a(an)
a. Interpreter b. Simulator
c. Compiler d. Commander

15. If DELHI is coded as 73541 and CALCUTTA is 82589662, how can CALICUT be coded?
a. 527943 b.5978213
c. 8251896 d. 8543691

16. In a certain code "256' means 'red colour halk','589' means'green colour flower' and '645' means 'white colour chalk', Which digit the code means 'white'?
a. 2 b. 4
c. 5 d. Cannot be determined

17. In a certain code language, 526 means blue,24 means blue colour and 436 means colour is fun. Which digit in that language means 'fun'?
a. 5 b. 4
c. 3 d. None of the above

18. Arun said, "This girl is the wife of the grandson of my mother". Who is Arun to the girl?
a. Father b. Grand father
c. Husband d. Father-in-law

19. Pesticide: Crop:: Antiseptic: ?

a. Wound b. Clotting
c. Bandage d. Bleeding

20. Cool: Cold

a. Length : Width
b. Gold : Bourne
c. Plant : Tree
d. Pretty : Beautiful

21.Choose the odd one out.

a. Pistol b. Sword
c. Gun d. Rifle

22. 13,32,24,43,35,.....46,65,57,76

a. 45 b. 52
c. 54 d. 55

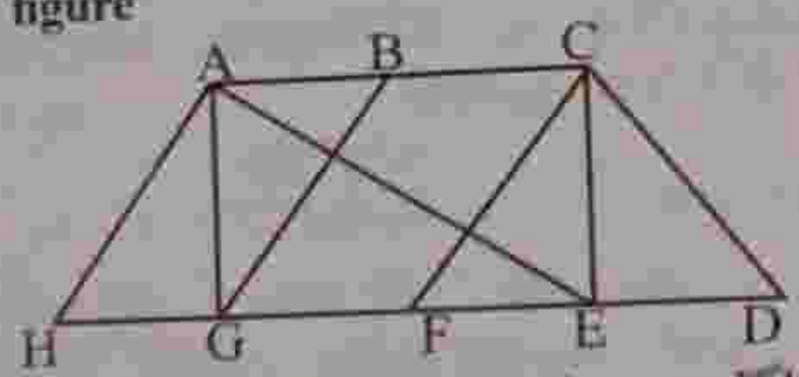
23. In a certain code, GIGANTIC is written as GIGTANCI. How is MIRACLES written in that code?

a. MIRLCAES b. MIRLACSE
c. RIMCALSE d. RIMLCAES

24. Y,W,U,S,Q,.....

a. N,J b. M,L
c. J,R d. L,M

25. Find the number of triangles in the given figure



a. 8 b.10 c. 12 d. 14

26. The angle of elevation of a ladder leaning against a wall is 60° and the foot of the ladder is 4.6 M away from the wall. the length of the ladder is

a. 2.3 M b. 4.6 M
c. 7.8 M d. 9.2 M

27. Which one will replace the question mark?

a. 41 b. 65
c. 35 d. 61

28. Which letter will replace the question mark?

a. Q b. A
c. Z d. K

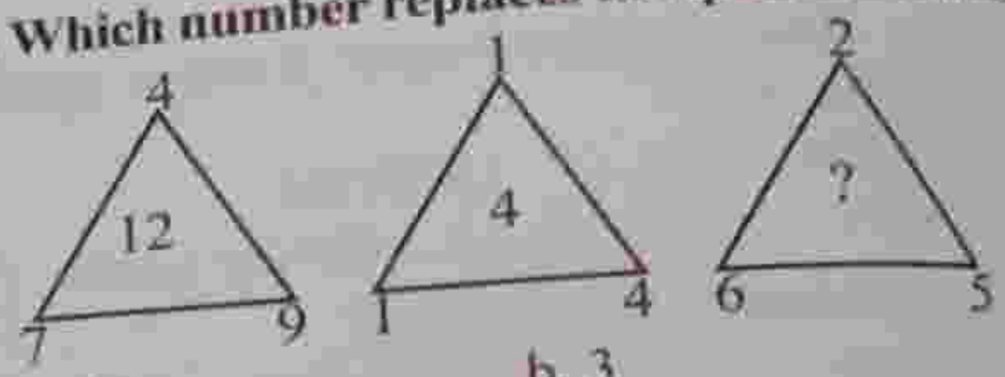
29. Which one of the following is not a prime number?

a. 31 b. 61
c. 71 d. 91

30. What least number must be added to 1056, so that the sum is completely divisible by 23?

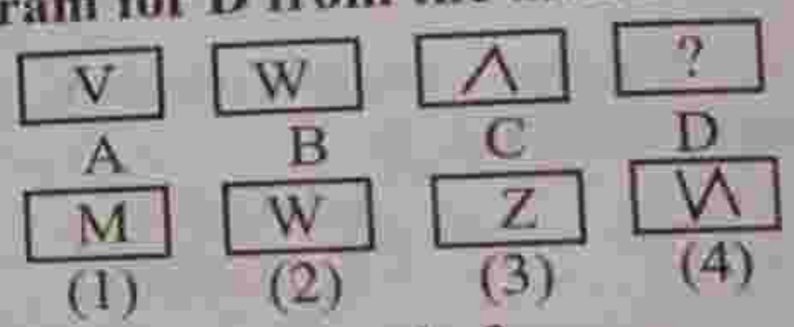
a. 2 b. 3
c. 18 d. 21

31. Which number replaces the question mark?



a. 15 b. 3
c. 9 d. 13

32. There is some relationship between diagrams A & B. The same relationship persists between C & D. Find the right diagram for D from the alternatives



a. 1 b. 2
c. 3 d. 4

33. Look at this series; 2, 1, $\left(\frac{1}{2}\right)$, $\left(\frac{1}{4}\right)$, ...

What number should come next?

a. $\left(\frac{1}{3}\right)$ b. $\left(\frac{1}{8}\right)$
c. $\left(\frac{2}{8}\right)$ d. $\left(\frac{1}{16}\right)$

34.Which word does NOT belong with others?

a. Inch b. Ounce
c. Centimetre d. Yard

35. Who out of the following was not a Moderate?

a. Gopal Krishna Gokhale

b. Dadabhai Naoroji
c. Ferozshah Mehta
d. Bal Gangadhar Tilak

36. **The minimum possible temperature beyond which matter cannot be cooled is**
a. -98.10°C b. -100.00°C
c. -273.15°C d. -459.40°C

37. **"Interest Rate Policy" is a component of**
a. Fiscal policy b. Monetary policy
c. Trade policy d. Direct control

38. **Which of the following was worshipped as a heavenly God by ancient Indians?**
a. Surya b. Angaraka c. Sukra d. Shani

39. **" A Vocie of Freedom" is a book written by**
a. Benazir Bhutto
b. Corazan Aquino
c. Aung San Sun Kyi
d. Nayantara Sahgal

40. **The Constitution of india borrowed the scheme of india Federation from the Constitution of .....**
a. USA b. Canada
c. Ireland d. UK

41. **The Russian traveller who visited the Deccan during the medieval period of Indian history was.....**
a. Rastyannikov b. Shirokov
c. Pavlor d. Nikitin

42. **Gerontology is the study of**
a. Process of ageing b. Growth of cells
c. Birds d. Vegetables

43. **Radio carbon dating process is used to know the age of**
a. Fossils b. Plants
c. Rocks d. All of the above

44. **The common ore of aluminium is**
a. Bauxite b. Chromite
c. Cryolite d. Monazite

45. **Choose the right synonym for 'RHAPSODY'**
a. Related to dance b. Related to printing
c. Related to music d. Related to acting

46. **Choose the right antonym for 'IMPROMPTY'**
a. Bad tempered b. Bad mannered
c. Irritable d. Well planned

47. **Fill in the blank with the correct verb: Neither Ravi nor his friends.....come.**
a. Has b. Have
c. Has been d. Have been

48. **Choose the right preposition for the following: The ball fell.....the well.**
a. by b. into
c. of d. at

49. **Choose the right choice of the words:**
Like in the rest of world, crime is ..... in Eastern Europe; but it has not reached alarming.....
a. serious....frequency
b. available...records
c. present.....proportions
d. unheard.....contours

50. **Choose the best analogy: Page-Margin**
a. Book-Print b. Building-Walls
c. City-Outskirts d. Road-Pavement

51. **Choose the right work.**
The dining hall should be very ..... to maintain large crowds
a. spacious b. specious
c. spesious d. spasious

52. **Use the appropriate forms of the word to fill in the blank.**
Soldiers for wars.....
a. brave b. braviestly
c. bravaly d. bravely

53. **Fill in the blank with an appropriate tense form of the verb.**
These dogs.....loudly.
a. Barking b. bark
c. barkly d. None of the above

54. **Choose the right passive form for the following: I know the man**
a. The man is known to me.
b. The man was known to me
c. The man knows me
d. The man is being known by me.

55. **Data (information) is stored in computer as**

a. Files b. Directories
c. Floppies d. Matter

56. Program stored in ROM are called...
a. Hardware b. Firmware
c. Software d. None of the above

57. ALU is
a. Arithmetic Logic Unit
b. Array Logic Unit
c. Application Logic Unit
d. None of the above

58. A source program is the program written in .....language.
a. Object b. Computer
c. Assembler d. High-level

59. The word length of a computer is measured in
a. Bytes b. Millimetres
c. Meters d. Bits

60. Which of the following is a storage device?
a. Tape b. Hard disk
c. Floppy disk d. All of the above

PART - 2

Each Question Carries Two Marks (20 x 2 = 40)

61. A parabola has its focus at (7, –4) and directrix y = 2, Find its equation
a. $x^2 + 12y - 14x + 61 = 0$
b. $x^2 - 14y + 12x + 61 = 0$
c. $x^2 - 12x + 14y + 61 = 0$
d. None of the above

62. How far from the x-axis is the focus F of the hyperbola $x^2 - 2y^2 + 4x + 4y + y = 0$?
a. 4.5 b. 3.4 c. 2.7 d. 2.1

63. The Eigen values of the Matrix $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 2 \\ 2 & 1 & 0 \end{bmatrix}$ **a**
a. 0, –1, 2 b. –1, 0, 1
c. 0, 1, 2 d. None of the above

64. The rank of the matrix will depend on
a. Number of rows of a matrix
b. Determinant of a matrix
c. Number of non-zero rows of a matrix
d. None of the above

65. The eigen values are $\begin{bmatrix} 1 & 1 \\ 2 & 5 \end{bmatrix}$ **are roots of the equation.**
a. $\lambda^2 - 6\lambda + 3 = 0$ b. $\lambda^2 - 6\lambda - 3 = 0$
c. $\lambda^2 + 6\lambda + 3 = 0$ d. None of the above

66. Find the distance between the points A(–4,7) and B(2,–5)
a. $8\sqrt{5}$ units b. $6\sqrt{5}$ units
c. $6\sqrt{4}$ units d. None of the above

67. The co-ordinates of end points of a diameter AB of a circle are A(–6,8) and B(–10,6). Find the co-ordinates of its center.
a. (–8, 7) b. (–7, –8)
c. (7, –8) d. (–8, –7)

68. Find the equation of a straight line whose slope is –2 which passes through the point (3, 7).
a. 2x + 2y – 13 = 0 b. 2x + y – 13 = 0
c. 2x + y + 13 = 0 d. 2x – y – 13 = 0

69. For what values of K, the following system is consistent?
x – 3y = k, 2x + 6y = –5
a. k = –5 b. k = 5
c. k = 0 d. k = –5/2

70. The system of equations
x – y + z = 1, x + y – z = 1, –x + y + z = 1 has
a. No solution b. Solution (1,1,1)
c. Solution (2,1,0) d. Many solutions

71. Is how many different ways can the letters of the word LEADING way that the vowls always come together?
a. 360 b. 480
c. 720 d. 5040

72. Out of 7 consonanta and 4 vowels, how many words of 3 consonants and 2 vowels can be formed?
a. 210 b. 1050
c. 25200 d. 21400

73. What is the value of b if $A = \begin{pmatrix} 1 & 4 \\ 2 & b \end{pmatrix}$ **is a singular matrix**
a. 5 b. 7 c. 6 d. 8

74. If the Pearson Correlation Coefficient R is equal to 1, then
a. there is a positive relationship between two variables
b. there is no relationship between two variables
c. there is a negative relationship between two variables
d. there is a perfect positive relationship between two variables

75. The coefficient of correlation between two variables X and Y is 0.8, while the regression coefficient of Y and X is 0.2. Then the regression coefficient of X an Y is
a. -3.2 b. 3.2
c. 2.3 d. -2.3

76. If the total sum of square is 20 and the sample variance is 5, then the total number of observations are
a. 15 b. 25
c. 4 d. 35

77. Suppose you buy a lottery ticket, you can either win one dollar (with probability 1/10) or win nothing (with probability 9/10). The amount you win is a random variable and it has a
a. Gamma distribution
b. Normal distribution
c. Exponential distribution
d. Bernoulli distribution

78. A firm estimates that the probability of having an employee off sick on any one day is 0.1. What is the probability that the firm experience six days without a worker off sick?
a. 0.001 b. 0.5
c. 0.531 d. .765

79. Where is the vertex of a parabola $X^2=4(y-2)^2$
a. (2,0) b. (0,2)
c. (3,0) d. (0,3)

Grade marks

80. What is the length of the length of the Latus Rectum of the curve $X^2=420y$?
a. $\sqrt{20}$ b. 20
c. 3 d. $\sqrt{5}$

ANSWERS

1 - a	2 - b	3 - a	4 - a	5 - d
6 - a	7 - c	8 - c	9 - c	10 - a
11 - b	12 - d	13 - d	14 - c	15 - c
16 - b	17 - c	18 - d	19 - a	20 - d
21 - b	22 - c	23 - b	24 - a	25 - d
26 - d	27 - b	28 - d	29 - d	30 - a
31 - c	32 - a	33 - b	34 - b	35 - d
36 - c	37 - b	38 - a	39 - d	40 - b
41 - d	42 - a	43 - a	44 - a	45 - c
46 - d	47 - b	48 - b	49 - c	50 - d
51 - a	52 - d	53 - b	54 - a	55 - a
56 - b	57 - a	58 - d	59 - a	60 - d
61 - a	62 - c	63 - a	64 - c	65 - a
66 - b	67 - a	68 - b	69 - d	70 - b
71 - c	72 - c	73 - d	74 - d	75 - b
76 - c	77 - d	78 - c	79 - G	80 - b